



PROJECT PROPOSAL

Sales & Customer Performance Dashboard

DEPI Final Project | Tableau

Retail Sales Data 2020–2023 | 3 Interactive Dashboards

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Field	Details
Project Title	Sales & Customer Performance — Tableau Dashboard
Client	American Retail Company
Data Period	2020 – 2023
Total Records	9,994 Orders
Dashboards	3 Interactive Dashboards: Sales Overview + Customer Analysis + Products Analysis
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Program	DEPI (Data Analytics Track)
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1. Project Overview

This project involves the design and development of three fully interactive, connected dashboards to analyze sales performance, customer behavior, and product analytics for an American retail company. The dashboards are built on four years of transactional data (2020–2023), covering 9,994 order records across three product categories: Furniture, Office Supplies, and Technology.

The initiative is driven by a user story from sales managers and executives who needed a clear, data-driven view of KPIs to make informed business decisions. All three dashboards are designed to be intuitive, visually consistent, and fully interactive — enabling stakeholders to slice and filter data by year, category, segment, and region, with seamless cross-dashboard navigation.

Background & Business Problem

The company lacked a unified analytics platform to monitor KPIs and make data-driven decisions despite having four years of transactional data and nearly 10,000 order records. Key gaps included:

- No unified view of sales, profit, and order KPIs across years
- Inability to compare current vs. prior year performance dynamically
- No visibility into which products, regions, or segments drive profit
- Customer behavior and retention trends were not being tracked
- Monthly and seasonal patterns were invisible to decision-makers
- Product performance gaps vs. sales targets were unmonitored

2. Objectives

The primary goal of this project is to empower sales managers and executives with actionable insights into business performance across three analytical dimensions:

Dashboard 1 — Sales Overview

- Provide a comprehensive high-level summary of Total Sales (\$733K), Total Profit (\$93K), Total Orders (1,687), and Growth Rate (+20.4%) for 2023
- Enable year-over-year comparison using a dynamic Select Year filter with previous year benchmarks
- Visualize monthly sales trends via a line chart showing seasonal momentum from \$20K (Feb) to \$118K (Nov)
- Identify top and bottom 3 products by sales using a horizontal bar chart
- Display quantity distribution across Office Supplies (7.68K), Furniture (2.44K), and Technology (2.36K)
- Show segment performance by region (Consumer, Corporate, Home Office) across Central, East, South, and West
- Present category distribution (Technology \$272K, Office Supplies \$246K, Furniture \$215K) via a donut chart
- Overlay monthly revenue area chart comparing current year vs. previous year performance

Dashboard 2 — Customer Analysis

- Show Total Customers (693, +8.6% YoY), Returning Customers (493, within-year buyers with 2+ orders), and Avg Orders per Customer (2.4, +18.1% YoY)
- Visualize monthly customer growth area chart (ranging from 53 in February to 216 in October)
- Analyze customer purchase behavior by segment (Consumer, Corporate, Home Office) across Sales, Profit, and Quantity
- Map customer concentration and revenue by US state (e.g., California \$146K, Texas - \$8.84K, Colorado -\$4.44K, Oregon -\$0.38K)
- Segment customers: 361 Consumer (52%), 204 Corporate (29%), 128 Home Office (18%) via donut chart
- Rank Top 10 customers by current year sales, profit, and quantity

Dashboard 3 — Products Analysis

- Highlight Top Product (Phones \$105K), Highest Revenue Category (Technology \$272K), and Average Profit per Order (\$28.21)
- Compare monthly sales vs. target using a combo bar-line chart; 5 months exceeded targets (Jan, Jun, Sep, Oct, Nov) and 4 months significantly missed (Feb, Mar, May, Dec)
- Display a revenue treemap breaking down categories and sub-categories
- Compare sales vs. profit across all 17 sub-categories using a dual horizontal bar chart

- Visualize monthly profit trend as a Waterfall/Gantt chart

3. Tools Used

Tool	Purpose	Details
Tableau Desktop	Dashboard Development	Used for all data connection, chart building, and dashboard layout
Power Query Editor	Data Cleaning & Transformation	Applied steps: Source, Promote Headers, Change Types, Remove Duplicates, Remove Blank Rows, Trim & Clean Text, Capitalize Each Word, Insert Year/Month columns
CSV Files (4 files)	Data Source	Semicolon-delimited, Latin-1 encoded. Contains Orders, Customers, Location, and Products tables
Star Schema Model	Data Modeling	Orders as the central Fact table linked to three Dimension tables via Customer ID, Postal Code, and Product ID
Calculated Fields / Measures	Business Logic	24+ custom calculated fields powering KPIs, sparklines, LOD expressions, and color logic
Parameters / Slicers	Dynamic Filtering	Select Year, Region, Category, and Segment filters drive all current-year vs. previous-year comparisons across all dashboards
Navigation Buttons	Cross-Dashboard UX	Interactive icon buttons on every page to navigate between all three dashboards smoothly
Microsoft Excel / CSV Viewer	Data Preview	Used to inspect and validate raw data before connecting in Tableau

4. Dataset Description

The dataset consists of four interconnected CSV files forming a Star Schema data model. Together they represent four years of retail operations for a US-based company selling across three major product categories.

Data Tables

Table	Type	Records	Key Columns
Orders	Fact Table	9,994 rows · 15 columns	Order ID, Customer ID, Product ID, Postal Code, Sales, Profit, Quantity, Discount, Ship Date, Segment
Customers	Dimension	793 rows · 2 columns	Customer ID, Customer Name
Location	Dimension	632 rows · 5 columns	City, State, Region, Country/Region, Postal Code
Products	Dimension	1,894 rows · 4 columns	Product ID, Category, Sub-Category, Product Name

Summary Statistics (2023 — Selected Year)

KPI	Value	YoY Change
Total Sales	\$733K	▲ +20.4% vs Pr Y
Total Profit	\$93K	▲ +14.2% vs Pr Y
Total Orders	1,687	▲ +28.3% vs Pr Y
Growth Rate	+20.4%	vs Previous Year
Total Customers	693	▲ +8.6% vs Pr Y
Returning Customers (within-year)	493	Customers with 2+ orders in 2023
Avg Orders / Customer	2.4	▲ +18.1% vs Pr Y
Top Product	Phones	\$105K in Sales

Data Cleaning & Transformation (Power Query Editor)

Table	Cleaning Steps Applied
Orders	Source → Promote Headers → Change Types → Remove Duplicates → Remove Blank Rows → Trimmed & Cleaned Text → Capitalize Each Word → Filtered Rows → Inserted Year/Month columns

Customers	Source → Promote Headers → Change Types → Remove Duplicates → Remove Blank Rows → Trimmed & Cleaned Text → Capitalize Each Word → Cleaned Text
Location	Source → Promote Headers → Change Types → Remove Duplicates → Remove Blank Rows → Trimmed & Cleaned Text → Capitalize Each Word → Filtered Rows → Cleaned Text
Products	Source → Promote Headers → Change Types → Remove Duplicates → Remove Blank Rows → Trimmed & Cleaned Text → Capitalize Each Word → Cleaned Text

5. Process

The project follows a structured 12-phase implementation process, moving from requirements analysis and data cleaning through to testing and final delivery.

#	Phase	Tasks	Duration
1	Requirements Analysis	Review user story, select chart types, sketch mockups for all 3 dashboards, define color palette	2–3 hrs
2	Data Cleaning & Preprocessing	Handle missing values, remove duplicates, standardize formats, fix incorrect data types, correct inconsistent values (via Power Query)	2–3 hrs
3	Data Source Setup	Connect 4 CSV files, build Star Schema, verify data types, dates, and numbers	1–2 hrs
4	Data Modeling & Relationships	Build table relationships, optimize data structure, create calculated columns and measures	2–3 hrs
5	Calculated Fields & KPIs	Year parameter + 18 sales/profit/qty fields + 6 customer fields + product metrics + field testing	3–4 hrs
6	Build Worksheets / Visuals	KPI cards, line/area/bar/donut/map/treemap charts across all 3 dashboards	5–7 hrs
7	Chart Formatting & UI Design	Remove gridlines, adjust axes, apply color logic, format numbers (K, \$), style tooltips, consistent typography	2–3 hrs
8	Dashboard 1 – Sales Overview	Build layout with 4 KPI cards, line chart, quantity bar, segment by region, donut, and area chart	3–4 hrs
9	Dashboard 2 – Customer Analysis	Customer KPIs, area growth chart, purchase behavior columns, US state map, segmentation donut, top 10 table	2–3 hrs

10	Dashboard 3 – Products Analysis	Build monthly sales vs target combo chart, treemap, dual bar chart, monthly profit waterfall/Gantt chart	2–3 hrs
11	Navigation & Interactivity	Add navigation icon buttons on every page, connect Year/Region/Category/Segment slicers, enable cross-dashboard sync	2–3 hrs
12	Testing & Delivery	Test all filters and interactivity, verify number accuracy, validate cross-dashboard sync, publish	1–2 hrs

Total Estimated Time: 27–40 hours

6. Dashboard Specifications

Dashboard 1 — Sales Overview

Filter Controls: Select Year, Category, Segment, Region (global filters synced across all visuals)

KPI Cards:

KPI	Value	YoY Change
Total Sales	\$733K	▲ +20.4% vs. Pr Y
Total Profit	\$93K	▲ +14.2% vs. Pr Y
Total Orders	1,687	▲ +28.3% vs. Pr Y
Growth Rate	+20.4%	vs. Previous Year

Charts:

- Line Sales Trend — Monthly sales from January (\$44K) through December (\$84K) with November peak at \$118K
- Quantity / Category — Vertical bar chart: Office Supplies 7.68K, Furniture 2.44K, Technology 2.36K
- Top & Bottom 3 Products — Horizontal bar chart with 17 sub-categories; Phones leading at the top
- Segment by Region — Stacked bar showing Consumer, Corporate, Home Office sales across Central, East, South, West
- Category Distribution Donut — Technology \$272K, Office Supplies \$246K, Furniture \$215K, Total \$733K
- Monthly Revenue (Current & Previous Year) — Dual-area overlay comparing current year vs. previous year monthly

Dashboard 2 — Customer Analysis

Filter Controls: Select Year, Region, Category, Segment

KPI Cards:

KPI	Value	YoY Change
Total Customers	693	▲ +8.6% vs. Pr Y
Returning Customers (within-year)	493	Customers with 2+ orders in 2023
Avg Orders per Customer	2.4	▲ +18.1% vs. Pr Y

Charts:

- Customer Growth Area Chart — Monthly trend from 53 (Feb) to peak 216 (Oct) showing business momentum

- Customer Purchase Behavior — Grouped columns for Consumer, Corporate, Home Office across Sales, Profit, and Quantity
- Customer Map — US state-level map showing customer concentration and revenue (California \$146K (+\$29K profit), Texas (-\$8.84K), Colorado (-\$4.44K), Oregon (-\$0.38K))
- Customer Segmentation Donut — 361 Consumer (52%), 204 Corporate (29%), 128 Home Office (18%), Total 693
- Top 10 Customers Table — Ranked by Current Year Sales, Profit, and Quantity; top customer Seth Vernon placed 73 orders

Dashboard 3 — Products Analysis

Filter Controls: Select Year, Region, Segment

KPI Cards:

KPI	Value	Note
Top Product	Phones	\$105K in sales
Highest Revenue Category	Technology	\$272K
Average Profit per Order	\$28.21	~6.5% margin on avg order value of ~\$434

Charts:

- Monthly Sales vs. Target — Combo bar-line chart; 5 months exceeded targets (Jan \$44K vs \$22K, Jun, Sep, Oct, Nov \$118K vs \$95K); 4 months significantly missed (Feb \$20K vs \$28K, Mar, May -\$24K gap, Dec \$84K vs \$116K)
- Revenue Treemap — Category → Sub-category breakdown: Technology (Phones \$105K, Copiers \$63K, Accessories \$60K), Office Supplies (Binders \$73K, Storage \$70K), Furniture (Chairs \$96K, Bookcases \$30K)
- Sales vs. Profit per Sub-Category — Dual horizontal bar across 17 sub-categories (Dark Navy = Sales, Teal = Profit)
- Monthly Profit Waterfall/Gantt Chart — Shows profit per month; Jan \$7.14K, Feb \$1.61K, Mar \$14.75K, Aug \$6.95K, Dec \$8.48K

7. Key Findings & Insights

Sales Performance (Dashboard 1)

- 2023 was the best-performing year with \$733K in sales and \$93K in profit, reflecting a +20.4% growth rate vs. prior year — all-time highs across every KPI
- Profit growth (+14.2%) lags order growth (+28.3%), signaling declining order value and potential discounting pressure
- Revenue swings from \$20K (February) to \$118K (November) — a 490% variance; extreme seasonality creates risk
- Monthly target performance is mixed: 5 months exceeded targets (January, June, September, October, November) and 4 months significantly missed (February, March, May, December). The May gap (-\$24K) is the single largest miss
- Technology is the dominant category at \$272K (37%), followed by Office Supplies \$246K (34%) and Furniture \$215K (29%)
- West Consumer (\$107K) is the single highest performing segment-region combination; South is weakest overall (~\$105K total)

Customer Behavior (Dashboard 2)

- 693 unique customers were active in 2023, with 493 returning customers (within-year buyers with 2+ orders), showing strong in-year loyalty
- Average orders per customer rose to 2.4 (+18.1%), indicating deeper engagement with the brand
- October was the peak customer acquisition month with 216 customers; February was the lowest at 53
- Consumer segment dominates at 361 customers (52%), followed by Corporate (204, 29%) and Home Office (128, 18%)
- California is the top revenue state at \$146K (+\$29K profit); Texas (-\$8.84K), Colorado (-\$4.44K), Oregon (-\$0.38K) are loss-making states
- Top customer Seth Vernon placed 73 orders; Grant Thornton generates \$8K sales but at -\$3.83K profit — a loss-making account
- Returning customers (493) represent strong in-year loyalty; new customer acquisition at +8.6% YoY needs acceleration

Product Performance (Dashboard 3)

- Phones lead all sub-categories at \$105K in sales, followed by Chairs (\$96K) and Binders (\$73K)
- 5 months exceeded sales targets and 4 months significantly missed; the largest miss is May (-\$24K gap) and the largest single-month shortfall is December (\$84K actual vs. \$116K target)
- Tables sub-category generates \$61K in sales but is profit-negative — actively destroying margin
- Copiers have the highest profit-to-sales ratio in the entire dataset — a strategic priority product

- Monthly profit peaked in March at \$14.75K; February saw only \$1.61K — confirming mid-year fragility
- Average profit of \$28.21/order on ~\$434 avg order value = ~6.5% margin — within range but needs improvement

8. Strategic Recommendations

All 10 recommendations are directly grounded in specific findings from the three interactive dashboards — enabling immediate, evidence-based action.

Part 1: Revenue & Growth

#	Recommendation	Trigger KPI	Target
01	Seasonal Sales Campaigns	February revenue (\$20K) is 83% below November peak; August actual (\$47K) misses its \$63K target — both recurring low-performance periods	Increase February revenue from \$20K to \$40K within 12 months; eliminate August target miss
02	Close the Target Gap	5 months exceeded targets; 4 months significantly missed in 2023. May gap (-\$24K) is the single largest miss. Q1/Q2 consistently underperforms	Achieve ≥90% of monthly target attainment within 2 quarters; eliminate May and February gaps first
03	Geographic Expansion — South & Central	West generates \$250K vs South's \$105K — 2.4× gap	Grow South + Central combined from \$251K to \$320K within 18 months
04	Resolve Negative-Profit States	Texas (-\$8.84K), Colorado (-\$4.44K), Oregon (-\$0.38K) lose money despite decent sales	Achieve breakeven (≥0 profit) in all loss-making states within 6 months

Part 2: Customer & Product

#	Recommendation	Trigger KPI	Target
05	Customer Loyalty & Retention Program	493 returning customers (within-year); avg orders/customer +18.1% to 2.4 orders — loyalty growing but not yet fully monetized	Sustain returning customer growth ≥30% YoY; grow avg orders per customer from 2.4 to 3.5 within 12 months; grow new customer acquisition to +15% quarterly
06	Eliminate Loss-Making Products & Accounts	Tables sub-category is profit-negative; Grant Thornton: \$8K sales but -\$3.83K profit — net loss customer	Convert all sub-categories to positive profit within 3 quarters; resolve Grant Thornton within 30 days

07	Discount Reduction & Margin Optimization	Average profit of \$28.21/order indicates thin 6.5% margins across the portfolio	Increase average profit per order from \$28.21 to \$35+ within 12 months
08	Cross-Sell Technology + Office Supplies Bundles	Technology leads in revenue but Office Supplies leads in quantity (7.68K units)	Increase average items per order by 25%; achieve 30% Technology accessory attach rate on Phone orders within 6 months

Part 3: Segmentation & Future Growth

#	Recommendation	Trigger KPI	Target
09	Customer Segmentation & Personalization	Consumer (52%), Corporate (29%), Home Office (18%) each have distinct needs; Corporate untapped in many regions	Grow Corporate segment from 29% to 35% over 24 months
10	Inventory Optimization for Peak Periods	November peaks at \$118K but other months average \$50–60K — seasonal success not replicated across the year	Increase November revenue to \$130K; grow March + June combined by 30%; achieve <2% stockout rate in Q4

9. Additional Features

- Modern and clean UI/UX design with a consistent navy-blue and teal color palette across all three dashboards
- Four interactive slicers (Year, Region, Category, Segment) synchronized across all dashboard pages
- Dynamic KPI cards showing current year values alongside YoY percentage change vs. prior year (with ▲/▼ directional arrows)
- Navigation icon buttons on every page to move smoothly between Sales Overview, Customer Analysis, and Products Analysis
- US state-level map visualization with customer count and revenue overlays in the Customer dashboard
- Revenue treemap with two-level drill-down (Category → Sub-Category) in the Products dashboard
- Top 10 Customers table with multi-metric ranking (Sales, Profit, Quantity) for relationship prioritization
- Dual-area monthly comparison chart (current vs. prior year) enabling instant trend spotting
- Sales vs. Target combo chart with actual vs. target line overlay for gap analysis
- Monthly Profit Waterfall/Gantt chart for intuitive profit distribution visualization
- Fully responsive and presentation-ready fixed layout at 1,200 × 800 px optimized for screen and export

10. Final Deliverables

Deliverable	Description	Format
Dashboard Sales Overview	4 KPI cards (Sales, Profit, Orders, Growth) + 6 interactive charts (line trend, quantity bar, top/bottom products, segment by region, category donut, monthly area) + global filters + navigation	.twbx
Dashboard Customer Analysis	3 KPI cards (Total Customers, Returning, Avg Orders) + 5 interactive charts (area growth, purchase behavior columns, US map, segmentation donut, top 10 table) + filters + navigation	.twbx
Dashboard Products Analysis	3 KPI cards (Top Product, Highest Revenue Category, Avg Profit) + 4 interactive charts (monthly sales vs target, treemap, dual bar, monthly profit waterfall) + filters + navigation	.twbx
Data Source	Cleaned Star Schema with all relationships, data type corrections via Power Query, and 24+ calculated fields / measures	Embedded
Filter / Slicer Panel	Cross-dashboard filters: Select Year, Region, Category, Segment — synced on all pages	Built-in
Navigation System	Icon navigation buttons on every page with consistent placement and active/inactive states	Built-in
Project Proposal	Full project documentation covering all phases, specs, key insights, and strategic recommendations	.pdf

Dashboard Specifications

Specification	Value
Dashboard Size	1,200 × 800 px — Fixed Size for consistent rendering
Primary Colors	Dark Navy (#1B3A6B) for headers, Light Blue (#D5E8F4) for alternating rows
Accent Colors	Teal/Cyan (#028090) for Consumer segment highlights and positive indicators; Dark Navy for Corporate
KPI Annotations	Year-over-year % change on every card with directional arrows (▲ positive / ▼ negative)
Interactivity	Dashboard Actions, Dynamic Tooltips, and Cross-Dashboard Navigation enabled on all charts

11. Business Impact, Challenges & Limitations

Business Impact

Impact Area	Description
Decision Speed	Managers can answer KPI questions in seconds vs. days of manual Excel analysis
Cost Visibility	Negative-profit states (TX, CO, OR) and sub-categories now visible — enabling \$17K–\$30K in immediate savings
Revenue Optimization	February dip and August slump identified — targeted campaigns can add \$50K+ in previously lost low-season revenue
Customer Intelligence	Top-10 table and segment map allow account managers to focus on high-value relationships and flag loss-making accounts like Grant Thornton
Target Accountability	Monthly gap tracking — with 5 outperforming and 4 underperforming months — creates a culture of data-driven performance management
Scalability	Star Schema and dynamic year parameter mean the same dashboards work for 2024 and beyond

Challenges & Limitations

Challenge	Description
Data Encoding	CSV files used Latin-1 encoding — required manual encoding conversion before import
Duplicate Records	Orders table contained duplicate entries requiring deduplication across 9,994 rows
No External Benchmarks	Analysis is self-referential — no industry benchmarks to validate growth rates against
Sales Target Source	Targets appear pre-set; no documentation of how targets were determined
Limited Demographics	No customer age, income, or behavioral data — limiting segmentation depth
4-Year Window	Only 2020–2023 data available — longer history would improve trend forecasting accuracy

12. Conclusion

This project delivers a complete, production-grade analytics solution tailored to the needs of a retail sales organization. By transforming 9,994 raw order records into three polished, fully interconnected interactive dashboards — Sales Overview, Customer Analysis, and Products Analysis — the project provides sales leadership with the visibility they need to drive growth, manage customer relationships, and make confident, data-backed decisions.

The complete pipeline — from data cleaning and preprocessing via Power Query, through Star Schema modeling and 24+ calculated fields, to three fully navigable dashboards — ensures that the solution is flexible, accurate, scalable, and easy to maintain. The consistent design language, including a unified navy-and-teal color palette, synchronized filter panels, dynamic KPI cards with YoY benchmarks, and cross-dashboard navigation, delivers a professional user experience that stakeholders can rely on day-to-day.

With an estimated delivery time of 27–40 working hours, the project is structured for efficient execution without compromising on quality or analytical depth.

The goal is not just to present data — it is to tell the complete story behind it: from revenue and growth, to customer loyalty and geography, to product-level profitability and target gaps. With 10 strategic recommendations directly grounded in dashboard findings, this solution is designed to generate immediate, measurable business value.

Expected Business Impact Summary

Recommendation Area	Expected Impact
Seasonal Sales Campaigns	+\$50K–\$80K annual revenue from February & August promotions
Discount Reduction & Margin Floor	+\$20K+ profit from pricing optimization
Loss-State Corrections & Grant Thornton repricing	+\$17K–\$30K from eliminating confirmed losses in TX (-\$8.84K), CO (-\$4.44K), OR (-\$0.38K) & Grant Thornton through pricing optimization
Customer Retention Program	5–10% increase in customer retention rate
February Trough Reduction	Reduce February from \$20K → \$40K (50% improvement)
Monthly Target Attainment	Achieve 90%+ monthly target attainment